ADEMP protocol: 06-component-decomposition

2026-05-07 16:50 PDT

This document is the working protocol for the exploratory simulation study that underlies `analysis/report/06-component-decomposition/`. The study is exploratory, and its findings should be confirmed in an independent study. ADEMP structure follows Morris, White and Crowther (2019).

Common infrastructure

Trial design. Hendrickson hybrid OL+BDC design with three phases: 8-week open-label active drug, 4-week blinded discontinuation, 4-week crossover. Within-participant assessment at the end of each phase plus weekly during the blinded phase.

Sample sizes. $N \in \{35, 70, 100, 150\}$, covering the trial-relevant range plus one larger anchor for asymptotic trend extrapolation.

Drug context. Prazosin for PTSD nightmares, calibrated to the Hendrickson 2020 reference parameters.

DGP architecture. Three-component additive generative model $Y_{it} = \mathrm{BL}_i - (BR_{it} + PB_{it} + TV_{it}) + \varepsilon_{it}$ with each component a participant-specific mod-Gompertz function and AR(1) within-participant residual correlation. The PB component includes the phase-dependent expectancy multiplier $\eta(\text{phase}) \in \{1.0, 0.5, 0\}$ for open-label-on-drug, blinded, and open-label-placebo phases respectively.

Replicates. $n_{\text{reps}} = 1000$ per cell for power and bias estimands; $n_{\text{reps}} = 5000$ for type I error and component-recovery cells where rare-event resolution is needed.

Random-number control. Per-replicate seed via cell-descriptor hash; base seed in `01-base-seed.txt`.

Reporting standard. Morris-White-Crowther ADEMP with MCSE columns for every reported performance measure.

Study A. Bias of one-component analysis under three-component DGP

A. Quantify the bias and Monte Carlo standard error of the biomarker-treatment interaction estimate when a one-component linear-mixed analysis is fitted to data generated under the three-component (BR-PB-TV) decomposition, as a function of the underlying PB and TV strengths.

D. Three-component DGP at the prazosin-PTSD reference m_{BR} values, with m_{PB} varied across $\{0, 1, 3, 6, 10\}$ and m_{TV} varied across $\{-1, 0, 1, 2\}$, generating a 5x4 grid of population-mean component compositions.

E. Primary estimand: $\hat{\beta}_{bm:D}$ from the analysis- model fixed-effects table. The bias is computed against the known biomarker-by-BR effect specified in the DGP.

M. Three analysis variants compared:

1. *One-component linear-mixed*: `Sx ~ bm + t + Dbc + bm:Dbc + (1 | ptID)` with `corCAR1(form = ~t|ptID)`.
2. *Phase-augmented linear-mixed*: `Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase + (1 | ptID)` with the same residual correlation.
3. *Full nonlinear three-component*: `nlme::nlme` fit of the modGompertz BR + PB + TV components with participant-specific random effects on each m. Fitted only at $N \geq 100$ where convergence rates exceed 80% in pilot.

P. Power for $H_0 : \beta_{bm:D} = 0$, type I error under $\beta_{bm:D} = 0$, bias of $\hat{\beta}_{bm:D}$, empirical SE across replicates, model SE, nominal-95% CI coverage, convergence rate. MCSE attached to every estimate.

Study B. Identifiability of the full decomposition

A. Characterise the identifiability of the participant- specific BR, PB, TV component estimates from the full nonlinear analysis at trial-relevant N.

D. As Study A, restricted to the $m_{PB} \in \{1, 6\}$ and $m_{TV} \in \{0, 1\}$ combinations.

E. Per-participant component estimates $(\hat{m}_{BR}, \hat{m}_{PB}, \hat{m}_{TV})$ and their joint posterior covariance.

M. Full nonlinear three-component fit only.

P. Convergence rate, mean fit time, condition number of the parameter-covariance matrix at convergence, and mean-squared error of each component estimate against the participant's true m value. Reported as a function of N to characterise the sample-size threshold for usable identifiability.

Study C. Subadditivity sensitivity (mechanistic vs emergent)

A. Quantify the bias in $\hat{\beta}_{bm:D}$ from the phase-augmented linear-mixed analysis under (i) a true mechanistic $BR \times PB$ subadditive interaction with coefficient γ, and (ii) a latent-class DGP with class-correlated BR and PB components but no direct interaction.

D. Two DGP families:

1. *Mechanistic subadditivity DGP*: $Y_{it} = \mathrm{BL}_i - [BR + PB + TV + \gamma \cdot BR \cdot PB] + \varepsilon$ with $\gamma \in \{-0.05, 0, 0.02, 0.05, 0.10\}$.
2. *Latent-class with class-correlated PB DGP*: as in the Study 1 mixture cell of paper 03 with `pb_class_correlation` axis varied.

E. $\hat{\beta}_{bm:D}$ from the phase-augmented analysis, plus $\hat{\gamma}$ from a saturated mean-structure analysis that includes the $BR \times PB$ interaction term explicitly.

M. Phase-augmented linear-mixed analysis (primary), saturated mean-structure analysis (secondary), class-aware `lcmm` analysis (tertiary, where applicable).

P. Power, bias, and MCSE under each DGP family. The falsifiability test: under the latent-class DGP with no true γ, the saturated mean-structure analysis should still recover non-zero $\hat{\gamma}$, demonstrating that $\hat{\gamma}$ alone is not evidence of mechanistic subadditivity.

Sequencing

1. **Phase 1: Study A.** Establishes the bias of the simple analysis as a function of population PB and TV strength.
2. **Phase 2: Study B.** Establishes the sample-size threshold for usable identifiability of the full decomposition.
3. **Phase 3: Study C.** Resolves the mechanistic-vs-emergent subadditivity question using the latent-class infrastructure from paper 03.

Deliverables

- Driver scripts at `analysis/scripts/component-decomposition/`.
- ADEMP results tables per study with MCSE columns alongside every estimate.
- Figures: bias-by-(PB strength, TV strength) heatmap; sample- size-by-identifiability curve; subadditivity sensitivity plot.

Protocol change log

`02-deviations-log.md` records changes to the cells, estimands, methods, or performance measures specified in this working protocol.

Source: `~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/analysis/scripts/component-decomposition/00-`